Chapter 14

Sets

Solutions

SECTION - A

Objective Type Questions (One option is correct)

[Sets and their Representation, Types of Sets, Cardinal Number, Equal set, Equivalent Sets, Power Set, Universal Set, Venn Diagram, Operations on Sets, De-Morgan's Law]

1. If $B = \left\{x : x = \dfrac{2n-1}{n+2}, n \in W \text{ and } n < 4\right\}$ then B can be written as

(1) $\left\{6, \dfrac{1}{3}, \dfrac{2}{4}, \dfrac{-1}{2}\right\}$ (2) $\left\{\dfrac{-1}{2}, \dfrac{1}{3}, \dfrac{3}{4}, 1\right\}$ (3) $\left\{3, \dfrac{-1}{2}, \dfrac{3}{4}, \dfrac{6}{5}\right\}$ (4) $\left\{\dfrac{1}{5}, \dfrac{1}{3}, \dfrac{-1}{2}, 1\right\}$

Sol. Answer (2)

Elements of B can be found by putting n = 0, 1, 2, 3 in $\dfrac{2n-1}{n+2}$

2. The set of the counting numbers which are multiples of 6 and less than 50 is written in set builder form as

(1) $\{x : x \in N$ and x is a multiple of 6$\}$ (2) $\{x : x = 6$ and $0 < x < 50\}$

(3) $\{x : x$ is a multiple of 6 and $0 < x < 50, x \in N\}$ (4) $\{x : x$ is a multiple of 6 and $x \leq 50\}$

Sol. Answer (3)

3. If A is the set of days in a week, then set A is

(1) Empty set (2) Singleton set (3) Finite set (4) Infinite set

Sol. Answer (3)

4. Which of the following is a null set?

(1) $\left\{x : x \in Z \text{ and } \dfrac{-1}{2} < x < \dfrac{9}{2}\right\}$ (2) $\{x : x \in N$ and $x - 4 \leq 15\}$

(3) $\{x : x \in N, x < 5$ and $x > 8\}$ (4) $\{x : x \in N$ and $x^2 < 40\}$

Sol. Answer (3)

5. Let $A = \{x : x$ is a positive multiple of 2 less than 20, $x \in N\}$, then $n(A)$ is

(1) 7 (2) 8 (3) 6 (4) 9

Sol. Answer (4)

$A = \{x : x$ is a positive multiple of 2, $0 < x < 20$, $x \in N\}$

$= \{2, 4, 6, 8, 10, 12, 14, 16, 18\}$

So $n(A) = 9$

6. If $P = \{x : x$ is a prime number less than 20$\}$ and $M = \{x : x$ is multiple of 6, $0 < x < 30\}$ then $n(P) - n(M)$ is

(1) 2 (2) 4 (3) 5 (4) 8

Sol. Answer (2)

$P = \{2, 3, 5, 7, 11, 13, 17, 19\} \Rightarrow n(P) = 8$

$M = \{6, 12, 18, 24\} \Rightarrow n(M) = 4$

$n(P) - n(M) = 8 - 4 = 4$

7. If $A = \{1, 2, 3\}$, then the number of elements in $P(A)$ is

(1) 6 (2) 8 (3) 12 (4) 16

Sol. Answer (2)

$n(A) = 3$

$n(P(A)) = 2^3 = 8.$

8. If $A = \{x, y\}$ then power set of A is

(1) $\{x^y, y^x\}$ (2) $\{\phi, x, y\}$ (3) $\{\phi, \{x\}, \{2y\}\}$ (4) $\{\phi, \{x\}, \{y\}, \{x, y\}\}$

Sol. Answer (4)

Power set is collection of all the subsets of given set.

9. Which of the following represents pair of equal sets?

(1) $A = \{1, 2, 3\}$, $B = \{x : x \in N$ and $x > 5\}$

(2) $A = \{$B, O, W, L$\}$, $B = \{x : x$ is a letter of the word ELBOW$\}$

(3) $A = \{1, 4, 9, 16\}$, $B = \{x : x = n^2, n \in N$ and $0 < n < 5\}$

(4) $A = \phi$, $B = \{$O, E, A$\}$

Sol. Answer (3)

$A = \{1, 4, 9, 16\}$

and $B = \{x : x = n^2$ and $0 < n < 5\}$

$= \{1, 4, 9, 16\}$

For all $x \in A$, $x \in B \Rightarrow A = B$.

10. If $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$, $C = \{1, 2, 3, 9, 6\}$ then which of the following is not true?

(1) $A \not\subset B$ (2) $A \subset C$ (3) $A \subset B$ (4) $B \not\subset C$

Sol. Answer (3)

$1 \in A$ but $1 \notin B$. So $A \not\subset B$.

11. Let A be the sets of the letters in the word "RAIGARH" and B be the sets of the letters in the word "PRATAPGARH" then $n(A \cap B)$ represents

(1) 7 (2) 6 (3) 4 (4) 3

Sol. Answer (3)

$A = \{R, A, I, G, H\}$

$B = \{P, R, A, T, G, H\}$

$A \cap B = \{R, A, G, H\} \Rightarrow n(A \cap B) = 4$.

12. If $D = \{x : x$ is divisible by 2 and 3 and $0 < x < 20\}$ and $B = \{x : x$ is a multiple of 6 and $0 < x < 25\}$ then $D - B$ is

(1) ϕ (2) $\{2, 3\}$ (3) $\{6, 12\}$ (4) $\{6, 12, 18\}$

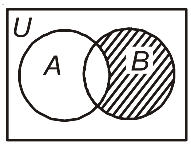
Sol. Answer (1)

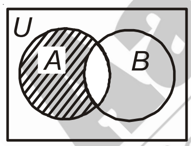
$D = \{6, 12, 18\}$

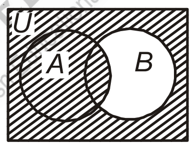
$B = \{6, 12, 18, 24\}$

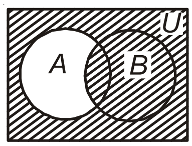
Then $D - B = \phi$

13. Which of the following Venn diagram represents $(A - B)$?

(1) 

(2) 

(3) 

(4) 

Sol. Answer (2)

$(A - B) = A - (A \cap B)$

14. For all the sets A and B, $A - (A \cap B)$ is equal to

(1) $A - (A \cup B)$ (2) $B - (A \cap B)$ (3) $A - B$ (4) $B - A$

Sol. Answer (3)

$A - (A \cap B) = A - B$.

15. Let $A = \{x : x$ is a prime number less than $10\}$ and $B = \{x : x \in N$ and $0 < x - 2 \leq 4\}$ then $A - B$ is

(1) $\{2, 3, 5\}$ (2) $\{3, 4\}$ (3) $\{3, 4, 5\}$ (4) $\{2, 7\}$

Sol. Answer (4)

$A = \{2, 3, 5, 7\}$

$B = \{3, 4, 5, 6\}$

Then $(A - B) = A - (A \cap B) = \{2, 3, 5, 7\} - \{3, 5\} = \{2, 7\}$.

16. If X and Y are two sets then $X \cap (X \cup Y)$ equals to

(1) X (2) Y (3) ϕ (4) $X \cap Y$

Sol. Answer (1)

$X \cap (X \cup Y) = X$.

17. If X and Y are two subsets of the universal set U and X' denotes the complement of X, then $X \cup (X \cap Y)'$ is equal to

(1) X (2) Y' (3) ϕ (4) U

Sol. Answer (4)

$X \cup (X \cap Y)' = X \cup (X' \cup Y')$ (De morgan's law)

$= (X \cup X') \cup Y'$ (Associative law)

$= U \cup Y'$ ($\because X \cup X' = U$)

$= U$

18. If $M = \{x : x \geq 7 \text{ and } x \in N\}$ for universal set of natural numbers, then M' is

(1) $\{1, 2, 3, 4, 5\}$ (2) $\{1, 2, 3, 4, 5, 6, 7\}$ (3) $\{1, 2, 3, 4, 5, 6\}$ (4) $\{0, 1, 2, 3, 4, 5, 6\}$

Sol. Answer (3)

$M = \{7, 8, 9, 10, \ldots\ldots\}$

then $M' = U - M$

$= N - M = \{x : x < 7, x \in N\}$

$= \{1, 2, 3, 4, 5, 6\}$

19. For any two sets A and B, $A' - B'$ is equal to

(1) $A - B$ (2) $B - A$ (3) $A - A'$ (4) $A - B'$

Sol. Answer (2)

$A' - B' = (U - A) - (U - B)$ ($\because X' = U - X$)

$= U - A - U + B$

$= B - A$

20. For any two sets A and B, the value of $[(A - B) \cup B]$ is equal to

(1) $A \cap B$ (2) $A \cup B$ (3) $A - B$ (4) $B - A$

Sol. Answer (2)

$(A - B) \cup B = (A \cap B') \cup B$

$= (A \cup B) \cap (B' \cup B)$

$= (A \cup B) \cap (U)$

$= A \cup B$

[Practical Problem on Union and Intersection, Cardinal Number of a Finite Set]

21. $n(A) = 20$, $n(B) = 18$ and $n(A \cap B) = 5$, then $n(A - B)$ is

(1) 13 (2) 12 (3) 15 (4) 18

Sol. Answer (3)

$n(A - B) = n(A \cup B) - n(B) = n(A) - n(A \cap B)$

$= 20 - 5$

$= 15$

22. If $n(U) = 50$, $n(A) = 20$, $n((A \cup B)') = 18$ then $n(B - A)$ is

(1) 14 (2) 12 (3) 16 (4) 20

Sol. Answer (1)

$n((A \cup B)') = 18$

$\Rightarrow n(U) - n(A \cup B) = 18$

$\Rightarrow n(A \cup B) = 32$

$n(B - A) = n(A \cup B) - n(A)$

$= 32 - 18$

$= 14$

23. There are 280 members in a club, each have at least one beverage. 100 of them drink tea and 75 drink tea but not coffee. Then the numbers of members drinking coffee is

(1) 100 (2) 150 (3) 105 (4) 205

Sol. Answer (4)

$n(T \cup C) = 280$, $n(T) = 100$

$n(T - C) = 75$

$\Rightarrow n(T \cup C) - n(C) = 75$

$\Rightarrow 280 - n(C) = 75$

$\Rightarrow n(C) = 280 - 75$

$\therefore n(C) = 205$

24. In a class of 60 students, 30 students like Mathematics, 25 like Science and 15 like both. Then, the number of students who like either Mathematics or Science is

(1) 30 (2) 40 (3) 45 (4) 50

Sol. Answer (2)

$n(U) = 60$

$m(M) = 30$, $n(S) = 25$, $n(M \cap S) = 15$

$n(M \cup S) = n(M) + n(S) - n(M \cap S)$

$= 30 + 25 - 15$

$= 40$

25. In a class of 50 students, 30 study Basic, 25 study Pascal and 10 study both. How many study neither?

(1) 15 (2) 0 (3) 5 (4) 35

Sol. Answer (3)

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$ $n(A) = 30$

$n(A \cup B) = 30 + 25 - 10$ $n(B) = 25$

$n(A \cup B) = 45$ $n(A \cap B) = 10$

$n(A' \cap B') = 50 - 45 = 5$

[Miscellaneous]

26. If a set has 2 elements then the difference between the subsets of its powers set and number of its subset is

(1) 2 (2) 8 (3) 10 (4) 12

Sol. Answer (4)

Number of subsets = $2^2 = 4$

Number of subsets of power = $2^4 = 16$

27. The set $A = \{x : a^x = 1, a > 0, x \in R\}$ can never be

(1) Null set (2) Singleton set (3) Finite set (4) Infinite set

Sol. Answer (1)

$a^x = 1 \Rightarrow x = 0, (a \neq 1)$

But if $a = 1$, then $x \in R$, Hence the set can never be null

28. The value of $(A \cup B \cup C) \cap (A \cap B^c \cap C^c) \cap C^c$ is

(1) $(B \cap C^c)$ (2) $(A \cap B^C \cap C^C)$ (3) $(B \cap C)$ (4) $(A \cap B \cap C)$

Sol. Answer (2)

Since $(A \cap B^C \cap C^C) \cap C^C = (A \cap B^C) \cap (C^C \cap C^C)$

$= A \cap B^C \cap C^C$

Since $A \cap B^C \cap C^C \subset A \cup B \cup C$

Therefore $(A \cup B \cup C) \cap (A \cap B^C \cap C^C) = A \cap B^C \cap C^C$

$(A \cup B \cup C) \cap (A \cap B^C \cap C^C) \cap C^C = A \cap B^C \cap C^C$

But this is not given therefore if $B^C \cap C^C \subset A$ then

$A \cap B^C \cap C^C = B^C \cap C^C$

29. If $A = \{1, 2, 3, 4\}$, $B = \{5, 6\}$ then the value of $A \Delta B$ is

(1) ϕ (2) $\{1, 2, 3, 4, 5, 6\}$ (3) $\{1, 2, 3\}$ (4) $\{5, 6\}$

Sol. Answer (2)

$A \Delta B = (A - B) \cup (B - A) = \{1, 2, 3, 4\} \cup \{5, 6\}$

$= \{1, 2, 3, 4, 5, 6\}$

30. If set A has 3 elements and set B has 6 elements then

(1) $6 \le n(A \cap B) \le 8$ (2) $6 \le n(A \cup B) \le 9$ (3) $0 \le n(A \cap B) \le 2$ (4) $1 \le n(A \cap B) \le 3$

Sol. Answer (2)

$n(A) = 3$, $n(B) = 6$

$\because \quad n(A \cup B) = n(A) + n(B) - n(A \cap B)$

If $n(A \cap B) = 0$ then

$n(A \cup B) = n(A) + n(B) = 3 + 6 = 9$

If $n(A \cap B) = 3$ because it may be that $A \subset B$

then $n(A \cup B) = 3 + 6 - 3 = 6$

therefore $6 \le n(A \cup B) \le 9$

and $0 \le n(A \cap B) \le 3$

31. A, B, C are non-empty sets then the value of $(A \cap B) \cap (B \cap C) \cap (C \cap A)$ is

(1) $A \cup B \cup C$ (2) $A \cup (B \cap C)$ (3) $A \cap B \cap C$ (4) $A \cap (B \cup C)$

Sol. Answer (3)

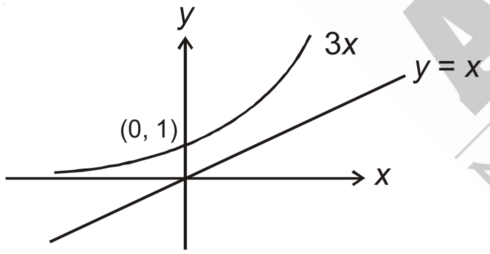
$(A \cap B) \cap (B \cap C) \cap (C \cap A) = (A \cap A) \cap (B \cap B) \cap (C \cap C)$

$= A \cap B \cap C$

32. If $A = \{(x, y) : y = 3^x, x, y \in R\}$ and $B = \{(x, y) : y = x, x, y \in R\}$, then which of the following is true?

(1) $B \subset A$ (2) $A \subset B$ (3) $A \cup B = A$ (4) $A \cap B = \phi$

Sol. Answer (4)



By the graph of $y = 3^x$ and $y = x$ it is clear that $A \cap B = \phi$

33. $A = \{\phi, \{\phi\}\}$, then $P(A)$ is

(1) $\{\phi, \{\phi\}\}$ (2) $\{\phi, \{\phi\}, \{\{\phi\}\}\}$

(3) $\{\phi, \{\phi\}, \{\{\phi\}\}, \{\phi, \{\phi\}\}\}$ (4) $\{\phi\}$

Sol. Answer (3)

$A = \{\phi, \{\phi\}\}$

Let $\{\phi\} = a$

$A = \{\phi, a\}$

$P(A) = \{\phi, \{\phi\}, \{a\}, \{\phi, a\}\}$

34. The number of sets X, such that $X \subseteq A$ and $X \not\subseteq B$, where $A = \{a, b, c, d, e\}$, $B = \{c, d\}$

(1) 26 (2) 27 (3) 28 (4) 29

Sol. Answer (3)

$X \subseteq A$ and $X \not\subseteq B$

$X \in P(A)$ and $X \notin P(B)$

Thus $X = \{a\}, \{b\}, \{c\}\ {}^5C_1 - 2$

$\{a, b\}, \{a, c\}$ ….. $= {}^5C_2 - 1$

$\{a, b, c\}, \{a, c, d\}$ ….. $= {}^5C_3$

$\{a, b, c, d\}, \{b, c, d, e\}$ ….. $= {}^5C_4$

$\{a, b, c, d, e\} = {}^5C_5$

Total $X = 28$

35. If $A = \phi$, $P(A)$ denotes power set of A, then number of elements in $P(P(P(P(P(A)))))$ is

(1) 1 (2) 2^4 (3) 2^5 (4) 2^{16}

Sol. Answer (4)

$n(P(A)) = 2^{\circ} = 1$

$n(P(P(A))) = 2^1 = 2$

$n(P(P(P(A)))) = 2^2 = 4$

$n(P(P(P(P(A))))) = 2^4 = 16$

$n(P(P(P(P(P(A)))))) = 2^{16}$

36. If A and B be two sets such that $n(A) = 15$, $n(B) = 25$, then number of possible values of $n(A \Delta B)$ (symmetric difference of A and B) is

(1) 30 (2) 16 (3) 26 (4) 40

Sol. Answer (2)

$n(A \Delta B) = n(A \cup B) - n(A \cap B)$

for maximum $n(A \Delta B)$, $n(A \cup B)$ should be maximum and $n(A \cap B)$ is minimum

For $n(A \cap B)$ to be minimum, $A \cap B = \phi$

$\Rightarrow$ $n(A \cup B) = 25 + 15 = 40$

$n(A \Delta B) = 40$

For $n(A \cap B)$ to be maximum, $A \subset B$

$n(A \cap B) = 15$

$\Rightarrow$ $n(A \cup B) = 25$

$\Rightarrow$ $n(A \Delta B) = 25 - 15 = 10$

$\Rightarrow$ Range of $(A \Delta B) = \{10, 12, 14, 16, \ldots, 40\}$

SECTION - B

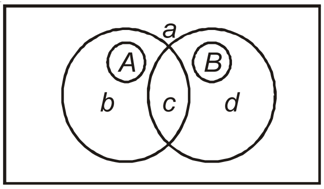
Objective Type Questions (More than one options are correct)

1. $|X|$ represent number of elements in region X. Now the following conditions are given

   $|U| = 14$, $|(A - B)^C| = 12$, $|A \cup B| = 9$ and $|A \Delta B| = 7$, where A and B are two subsets of the universal set U and A^C represents complement of set A, then

   (1) $|A| = 2$ (2) $|B| = 5$ (3) $|A| = 4$ (4) $|B| = 7$

Sol. Answer (3, 4)



$a + b + c + d = 14$ …(i)

$a + c + d = 12$ …(ii)

$b + c + d = 9$ …(iii)

$b + d = 7$

$b = 2,\ a = 5,\ d = 5,\ c = 2$

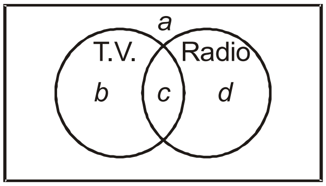
$|A| = b + c = 4$

$|B| = d + c = 7$

2. Out of 14 people, twelve said that it was not the case that they watched television but did not listen to the radio. Also, for nine people it is not the case that they do not watch T.V. and do not listen to the radio. Finally, seven people either watch television or listen to the radio but do not do both. Now let A be number of people that watch T.V. and B be the number of people that listen radio. Then

   (1) $A = 2$ (2) $B = 5$ (3) $A = 4$ (4) $B = 7$

Sol. Answer (3, 4)



$a + b + c + d = 14$ …(i)

$a + c + d = 12$ …(ii)

$b + c + d = 9$ …(iii)

$b + d = 7$ …(iv)

$a = 5,\ b = 2,\ c = 2,\ d = 5$

$|A| = b + c = 4$

$|B| = c + d = 7$

3. If $X = \{1, 2, 3, 4\}$, $Y = \{2, 3, 5, 7\}$, $Z = \{3, 6, 8, 9\}$, $W = \{2, 4, 8, 10\}$, then

(1) $(X \Delta Y) \Delta (Z \Delta W)$ is $\{1, 2, 3, 5, 6, 7, 9, 10\}$

(2) $(X \Delta Z) \Delta (Y \Delta W)$ is $\{1, 2, 3, 5, 6, 7, 9, 10\}$

(3) $(X \Delta Y) \Delta (Y \Delta W)$ is $\{1, 3, 8, 10\}$

(4) $(X \Delta Y) \Delta (Z \Delta W)$ is $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Sol. Answer (1, 2, 3)

$X = \{1, 2, 3, 4\}$, $Y = \{2, 3, 5, 7\}$, $Z = \{3, 6, 8, 9\}$, $W = \{2, 4, 8, 10\}$

(1) $(X \Delta Y) \Delta (Z \Delta W)$

Let $K_1 = (X \Delta Y) = (X - Y) \cup (Y - X) = \{1, 4, 5, 7\}$

$K_2 = (Z - W) \cup (W - Z) = \{3, 6, 9, 2, 4, 10\}$

Now, $K_1 \Delta K_2 = (K_1 - K_2) \cup (K_2 - K_1)$

$\Rightarrow \quad \{1, 5, 7, 3, 6, 9, 2, 10\} = \{1, 2, 3, 5, 7, 9, 10\}$

Similarly by applying

$(A \Delta B) = (A - B) \cup (B - A)$

Option (1), (2), and (3) are correct but not (4).

4. For any three sets. P, Q and R, S is an element of $(P \Delta Q) \Delta R$, then S may belong to

(1) Exactly one of P, Q and R, if P, Q and R disjoint sets

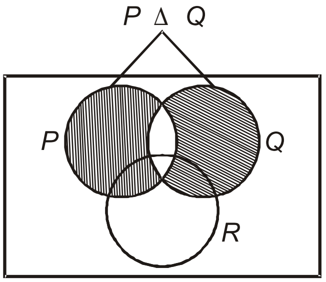
(2) At least one of P, Q and R, but not in all three of them at the same time

(3) Exactly two of P, Q and R

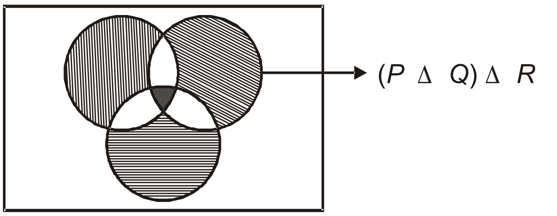
(4) Exactly one of P, Q and R or in all the three of them

Sol. Answer (1, 4)

P, Q, R



$(P \Delta Q) \Delta R$



5. For a set {1, 2, {1, 2, 3}}. Which of the following statement is false?

(1) $\{1, 2\} \in \{1, 2, \{1, 2, 3\}\}$ (2) $\{1, 2\} \subseteq \{1, 2, \{1, 2, 3\}\}$

(3) $\{1, 2, 3\} \subseteq \{1, 2, \{1, 2, 3\}\}$ (4) $3 \in \{1, 2, \{1, 2, 3\}\}$

Sol. Answer (1, 3, 4)

By using properties of sets and subset only option (2) is correct.

SECTION - C

Linked Comprehension Type Question

Comprehension

In a certain city of 15000 families, 3.5% of families who read *A* but not *B* look into advertisements, 25% of the families who read *B* but not *A* look into advertisements and 50% of the families, who read both *A* and *B* look into advertisements. It is known that 8000 families read *A*, 4000 read *B* and 1000 read both *A* and *B*.

1. The number of families who look into advertisements

(1) 1295 (2) 1395 (3) 1495 (4) 1500

Sol. Answer (3)

$n(A \cap \bar{B}) = 8000 - 1000 = 7000$

$n(\bar{A} \cap B) = 4000 - 1000 = 3000$

$n(A \cap B) = 1000$

Required number of persons $= \frac{35}{100} \times 7000 + \frac{25}{100} \times 3000 + \frac{50}{100} \times 1000$

$= 1495$

2. The number of families who read none of the newspaper

(1) 3000 (2) 4000 (3) 6000 (4) No family

Sol. Answer (2)

15000 – [8000 + 4000 – 1000] = 4000

3. The number of families who read atmost one of the newspaper

(1) 11000 (2) 12000 (3) 13000 (4) 14000

Sol. Answer (4)

$A(\bar{A} \cap \bar{B}) + n(\bar{A} \cap B) + n(A \cap \bar{B})$

4000 + 7000 + 3000 = 14000

SECTION - D

Assertion-Reason Type Question

1. STATEMENT-1 : If $A = \{x \mid x \in R, x \geq 2\}$ and $B = \{x \mid x \in R, x < 4\}$ then $A \Delta B = R–[2, 4)$.

and

STATEMENT-2 : For any two sets A and B, $A \Delta B = (A – B) \cup (B – A)$.

Sol. Answer (1)

$A\Delta B = (A \cup B) – (A \cap B)$

Hence in this case $A\Delta B = R – [2, 4)$

SECTION - E

Matrix-Match Type Questions

1. $A = \{x : x \in N, \text{G.C.D.}(x, 36) = 1, x < 36\}$, $B = \{y : y \in N, \text{G.C.D.}(y, 40) = 1, y < 40\}$; (G.C.D. stands for greatest common divisors)

Column-I		**Column-II**
(A)	$n(A \cap B)$	(p) 10
(B)	$C = \{x : x \in A \cup B, x \text{ is prime}\}, n(C) =$	(q) 9
(C)	$n(A \Delta B)$	(r) 21
(D)	$n((A – B) \times (B – A))$	(s) 11

Sol. Answer A(q), B(s), C(p), D(r)

$A = \{1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35\}$

$B = \{1, 3, 7, 9, 11, 13, 17, 19, 21, 23, 27, 29, 31, 33, 37, 39\}$

$x(A \cap B) = 9$

$x(A \Delta B) = 10$

$C = \{x : x \in A \cup B, x \text{ is prime}\}$

$= \{3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37\}$

$x(C) = 11$

$x((A – B) \times (B – A)) = x(A – B)\cdot x(B – A) = 3 \times 7 = 21$

SECTION - F

Integer Answer Type Questions

1. Let $X = \{1, 2, 3,.....10\}$ and $P = \{1, 2, 3, 4, 5\}$. The number of subsets Q of X such that $P \Delta Q = \{3\}$ is _____.

Sol. Answer (1)

For $P \Delta Q = \{3\}$

Element 3 should be either in P or in Q.

As it is given in P must not in Q and all remaining elements should be same so only 1 way is possible.

2. Two finite set have m and n elements. The total number of subsets of the first set is 56 more than the total number of subsets of second set. Then $|m - n|$ is equal to _____.

Sol. Answer (3)

$2^m - 2^n = 56 \Rightarrow 8 \times 7 = 2^3 \times 7$

$2^n(2^{m-n} - 1) = 2^3 \times 7 \Rightarrow n = 3 \quad 2^{m-n} = 8 = 2^3 = \boxed{m-n=3}$

3. $A = \{x : x$ satisfies $x^3 - 11x^2 + 39x - 45 = 0\}$

$B = \{x : 5x - 6, 3x + 1, x - 1$ are lengths of sides of a $\Delta\}$, then $n(A \cap B)$ is _____.

Sol. Answer (2)

$x^3 - 11x^2 + 39x - 45 = 0$

$\Rightarrow (x - 3)(x^2 - 8x + 15) = (x - 3)(x - 3)(x - 5) = 0$

$\Rightarrow x = 3, 5$

$5x - 6 + 3x + 1 > x - 1$

$7x > 4 \Rightarrow x > \frac{4}{7}$

$5x - 6 + x - 1 > 3x + 1$

$3x > 8 \Rightarrow x > \frac{8}{3}$

$3x + 1 + x - 1 > 5x - 6$

$\Rightarrow x < 6$

$\Rightarrow \frac{8}{3} < x < 6$

4. A, B, C be three sets such that $n(A) = 2$, $n(B) = 3$, $n(C) = 4$. If $P(x)$ denotes power set of X,

$K = \frac{n(P(P(C)))}{n(P(P(A))) \times n(P(P(B)))}$. Sum of digits of K is _____.

Sol. Answer (7)

$$K = \frac{2^{2^4}}{2^{2^2} \times 2^{2^3}} = \frac{2^{16}}{2^4 \times 2^8} = 2^4 = 16$$

SECTION - G

Multiple True-False Type Question

1. STATEMENT-1 : Number of elements belonging to exactly two of the sets A, B, C is

$$n(A \cap B) + n(B \cap C) + n(C \cap A) - 3n(A \cap B \cap C)$$

STATEMENT-2 : Number of elements belonging to atmost two of sets A, B and C is

$$n(A \cup B \cup C) - n(A \cap B \cap C)$$

STATEMENT-3 : Number of elements belonging to exactly one of the sets A, B and C is

$$n(A \cup B \cup C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + 2n(A \cap B \cap C)$$

(1) T T T (2) T T F (3) T F T (4) F T T

Sol. Answer (1)

All are true. Standard Results.

SECTION - H

Aakash Challengers Questions

1. Let $A_1, A_2, \ldots\ldots, A_m$ be m sets such that $O(A_i) = p \ \forall \ i = 1, 2, \ldots\ldots, m$ and $B_1, B_2, \ldots\ldots, B_n$ be n sets such that $O(B_j) = q \ \forall \ j = 1, 2, \ldots\ldots, n$. If $\bigcup\limits_{i=1}^{m} A_i = \bigcup\limits_{j=1}^{n} B_j = S$ and each element of S belongs to exactly α number of A_i's and β number of B_j's, then

(1) $pm = nq$ (2) $\alpha pm = \beta nq$ (3) $\beta pm = \alpha nq$ (4) $(pm)^{\beta} = (nq)^{\alpha}$

Sol. Answer (3)

$\sum\limits_{i=1}^{m} O(A_i) = mp$, in which each element of S is added α number of times, hence

$$O(S) = \frac{mp}{\alpha}$$

Similarly, $O(S) = \dfrac{nq}{\beta}$

$\Rightarrow \quad \dfrac{mp}{\alpha} = \dfrac{nq}{\beta}$

$\Rightarrow \quad mp\beta = nq\alpha$

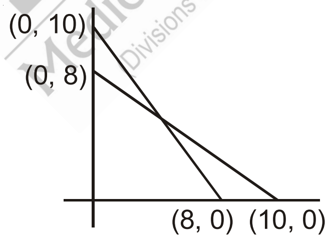
2. $A = \{(x, y) : x, y \in I, x \geq 0, y \geq 0 \text{ and } 4x + 5y \leq 40\}$

$B = \{(x, y) : x, y \in I, x \geq 0, y \geq 0 \text{ and } 5x + 4y \leq 40\}$

where I denotes set of integers, then $n(A \cap B) =$

(1) 45 (2) 36 (3) 55 (4) None of these

Sol. Answer (1)



x	y	n
0	0, 1, 2,....., 8	9
1	0, 1,....., 7	8
2	0, 1, 2,....., 6	7
8	0	1

Total $= \dfrac{9 \times 10}{2} = 45$

3. In a battle, 70% of the combatants lost one eye, 80% lost one leg, 85% lost ear, 90% an arm. If x% lost all the four organs, then minimum value of x

(1) 25% (2) 20% (3) 30% (4) 35%

Sol. Answer (1)

Paragraph (Q.4 to Q.6)

Times , Mirror and Sun are three newspapers.

(i) All readers of the Times read the Sun.

(ii) Every person either reads the Sun or does not read the Mirror.

(iii) 11 people read the Sun but does not read the Mirror.

(iv) 8 people read either the Times or the Mirror but not both.

(v) 10 people read the Sun and either read the Mirror or do not read the Times.

(vi) 14 people either read the Sun and not the Mirror or read both the Sun and Times.

(vii) 9 people neither read the Times nor the Mirror.

4. Number of people who read only sun are

(1) 6 (2) 5

(3) 16 (4) 18

5. Number of people who read Times and Mirror both are

(1) 3 (2) 2

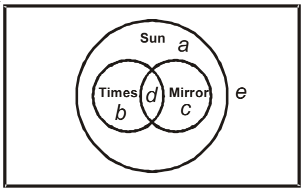
(3) 4 (4) 5

6. Number of people who do not read any of the newspaper

(1) 0 (2) 2

(3) 4 (4) 3

Solution for (Q.4 to Q.6)



Now, we know from first two statements times and mirror are subsets of Sun.

Now,

According to statement-3, $a + b = 11$ …(i)

According to statement-4, $b + c = 8$ …(ii)

According to statement-5, $a + c + d = 10$ …(iii)

According to statement-6, $a + b + d = 14$ …(iv)

According to statement-7, $a + e = 9$ …(v)

Solving these

$a = 5$, $b = 6$, $c = 2$, $d = 3$, $e = 4$

4. Answer (2)

5. Answer (1)

6. Answer (3)

7. If A and B be two sets such that $n(A) = 15$, $n(B) = 25$, then number of elements in the range of $n(A \Delta B)$ (symmetric difference of A and B) is

(1) 31 (2) 16

(3) 26 (4) 40

Sol. Answer (2)

$n(A \Delta B) = n(A \cup B) - n(A \cap B)$

for maximum $n(A \Delta B)$, $n(A \cup B)$ should be maximum and $n(A \cap B)$ is minimum

For $n(A \cap B)$ to be minimum, $A \cap B = \phi$

$\Rightarrow$ $n(A \cup B) = 25 + 15 = 40$

$n(A \Delta B) = 40$

For $n(A \cap B)$ to be maximum, $A \subset B$

$n(A \cap B) = 15$

$\Rightarrow$ $n(A \cup B) = 25$

$\Rightarrow$ $n(A \Delta B) = 25 - 15 = 10$

$\Rightarrow$ Range of $(A \Delta B) = \{10, 12, 14, 16, \ldots, 40\}$

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